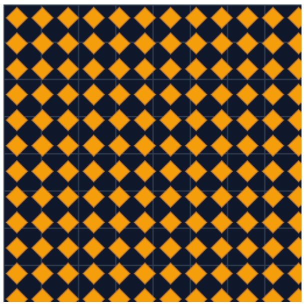
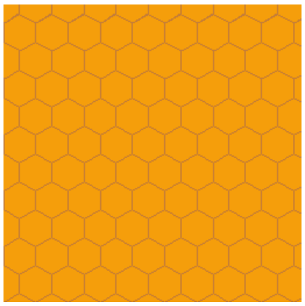
Texture Library Tutorial

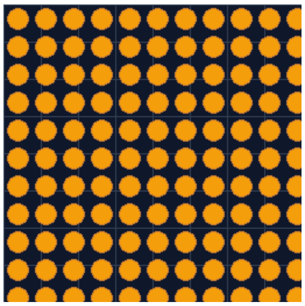
Designing printable surfaces with knurls, hex grids, diamond plate, fabric weave, voronoi — and how to wrap them onto cylinders.



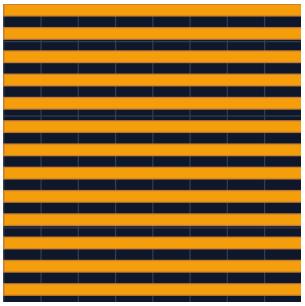
Knurl (Diamond)



Hex grid

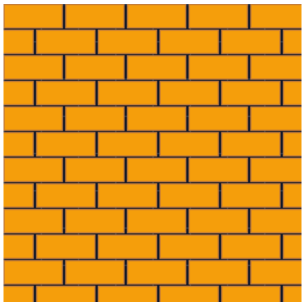


Bumps

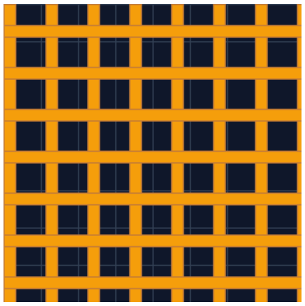


Ridges (linear)

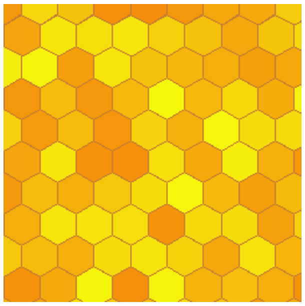
Diamond plate



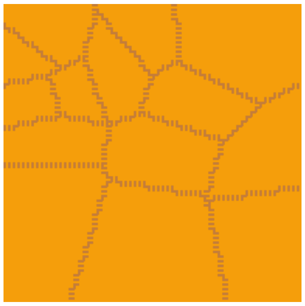
Brick wall



Fabric weave



Hex camo



Voronoi

All nine textures produce **real, printable surface geometry** — they survive STL export, CSG operations and slicing. They are not visual-only image maps.

1. Why geometric (not image) textures?

Most CAD packages and renderers fake textures with images projected onto a surface. Those textures look right on screen but they don't exist in 3D — when the slicer produces G-code, the model is just smooth plastic again. Useless for printing.

ForgeSlicer's textures are **actual displaced surface geometry**: every bump, groove, hex cell, and voronoi wall is a tessellated 3D shape. When the model gets sliced, those reliefs become real tool-path movements. The result prints exactly as drawn.

Trade-off: more triangles. A 60×60 mm bumps panel can carry 8–15 k triangles by itself; dense knurl on a flashlight grip can hit 40–60 k. The renderer and slicer both handle these comfortably, but if you're authoring on a very low-end device, prefer larger *tile size* values to drop count fast.

Modifier: positive vs. negative

Every texture object behaves like any other primitive — it has a **modifier** flag. The flag decides whether the relief is added to or carved from the host:

- **Positive** — the texture is union'd onto the host part. Use this for raised reliefs: tool-handle grips, diamond-plate flooring, decorative panels.
- **Negative** — the texture is subtracted from the host. Use this for engraved logos, serial numbers, drainage channels, gripping treads.

Switch the modifier from the Inspector's **+/**– toggle at any time — the underlying mesh doesn't change, only the CSG operation applied to it.

2. Adding a texture

Method A — From the Texture Library dialog

Open the workspace, click **Library** → **Textures** in the left toolbar (or use the keyboard shortcut **Shift+T**). The dialog shows all nine patterns with live preview thumbnails. Pick one, tune the sliders, hit **Add to scene**. A new texture object lands at the world origin, ready to position.

Method B — Right-click an existing face

Right-click any visible primitive in the scene and choose **Apply texture to face…**. The Texture Library dialog opens, but now you'll see a *face picker* at the top — choose *+X / –X / +Y / –Y / +Z / –Z* and the texture's width/depth are auto-sized to that face's bounding box. The new texture object inherits the target's transform so it sits flush against the picked face.

Tip: set the modifier to **Negative** in the picker before adding if you want to engrave rather than emboss. You can flip it later in the Inspector but it saves a round-trip.

Method C — Voice command

If you have voice mode enabled, say *“add bumps texture, three by three centimeters, height one millimeter”*. The Whisper transcription is parsed by GPT-5.2 into a texture-creation intent — same parameters as the dialog, just faster when your hands are on the model.

3. Tuning a texture: the four key parameters

Parameter	Effect	Typical range	Notes
w, d (footprint)	Width × depth of the textured region	10–120 mm	Auto-sized when applying via right-click face picker.
tileSize	Periodicity of one tile in mm	1.5–6 mm	Smaller = denser (more triangles, finer feel).
height	Depth of the relief	0.4–3 mm	For positive: how far it sticks up. For negative: how deep it
depth (base)	Thickness of the base plate	0.4–1.5 mm	The texture sits ON TOP of a thin base — keep ≥ 0.4 mm f

Triangle budget rule of thumb: a 50×50 mm patch at tileSize=2 mm produces 20–30 k triangles for the dense patterns (knurl, hex). Doubling tileSize drops the count by roughly 4× without changing the visual character.

4. Wrap modes — flat vs. cylinder

By default a texture is a flat panel sitting on the XZ plane. The Texture Library dialog has a **Wrap** selector that lets you bend the flat tile into a **cylinder**, with a configurable radius. This is the single most useful feature for things like flashlight grips, dial-knurls, vase patterns, and decorative columns.

Two wrap modes:

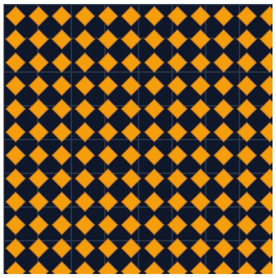
- **Flat** — texture stays planar (the default).
- **Cylinder** — width *w* wraps around the cylinder's circumference. With **Auto radius** (*wrapRadius=0*) the dialog picks a radius so the texture wraps *exactly once* around the cylinder. Set a manual radius to control how tight the wrap is.

Maths: auto radius = $w \div 2\pi$. So a 60 mm wide texture on auto radius produces a cylinder of ~9.5 mm radius. To wrap onto a known cylinder, set $w = 2\pi \times$target radius (or just enter the radius directly).

Sphere wrap? Not yet supported. A flat-to-sphere mapping requires non-uniform tile scaling near the poles to avoid pinching — on the V2 backlog.

5. Pattern catalog — pick the right one

5.1 Knurl (Diamond)

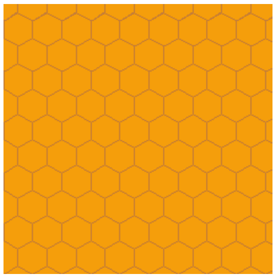


Diagonal cross-hatch of square pyramids. The grip standard.

Use cases. Tool handles, dial-knurls, anti-twist surfaces on threaded caps.

Tip. tileSize 1.5–2.5 mm gives a fine, fingertip-grippable feel. Larger tiles (4 mm+) look like aggressive industrial knurls.

5.2 Hex grid

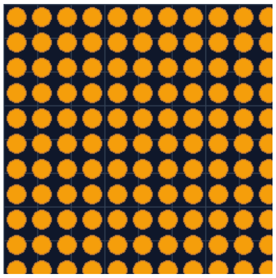


Tiled hexagonal cells with flat tops.

Use cases. Vents, honeycomb panels, lightweight infill plates, decorative.

Tip. Use as a NEGATIVE on a thin plate to produce a perforated grille. Combine with a positive base to control structural strength.

5.3 Bumps

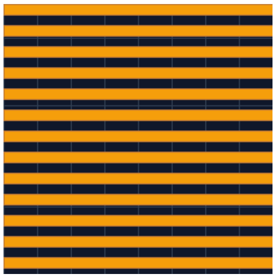


Hemispherical dots on a regular grid.

Use cases. Anti-slip pads, dot patterns, tactile indicators, braille-style.

Tip. Height 0.8 mm with tileSize 3 mm is the standard rubber-foot feel. For braille, tileSize 2.5 mm + height 0.5 mm hits the spec.

5.4 Ridges (linear)



Parallel half-cylinder grooves.

Use cases. Flashlight grip, column fluting, threaded-cap finger ridges.

Tip. Apply with **Cylinder wrap** to get classic vertical fluting around a cylindrical body. Orient the texture's rotation to point the ridges along whichever axis you want.

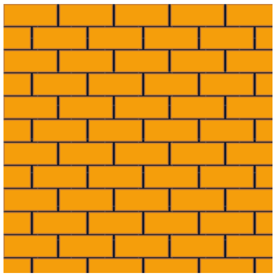
5.5 Diamond plate

Pinwheel of four diamonds per tile — industrial tread.

Use cases. Floor panels, stair treads, slip-resistant covers.

Tip. Default height 1.5 mm is correct — anything less reads as wallpaper. Don't shrink tileSize below 4 mm; the pinwheel detail collapses.

5.6 Brick wall

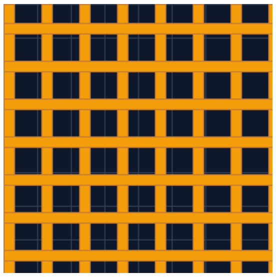


Running-bond brick layout with mortar gaps.

Use cases. Decorative wall panels, dollhouse facades, miniature dioramas.

Tip. Use as POSITIVE on a thin baseplate; the mortar gaps become the low-relief 'in-between' surface. Default tileSize is the brick width.

5.7 Fabric weave

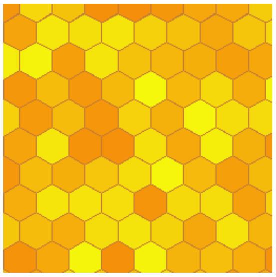


Basket-weave warp+weft cylinders.

Use cases. Burlap, canvas, woven-mat insets, decorative coasters.

Tip. Looks best at tileSize 3–4 mm with height ~ tileSize/3. Cylinder wrap turns this into a woven-grip sleeve.

5.8 Hex camo

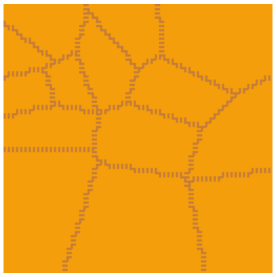


Hex grid with randomised cell heights.

Use cases. Military-style armor panels, futuristic skins, gaming props.

Tip. The height variation is per-cell deterministic — same seed → same panel. Save your scene; the random heights persist across reloads.

5.9 Voronoi



Irregular polygonal cells from random seed points.

Use cases. Cracked-glass effects, organic stipple, natural-looking texturing.

Tip. Best as a NEGATIVE on a flat panel: the cell walls become engraved channels that catch light beautifully under FDM line-art lighting.

6. CSG workflow — how textures combine with parts

Texture objects participate in ForgeSlicer's CSG (Constructive Solid Geometry) engine just like any other primitive. The order of operations is:

1. All **positive** primitives in the scene are union'd together to form the base solid.
2. All **negative** primitives are then subtracted from that base solid.
3. The final mesh is what gets sliced and exported as STL / 3MF.

Practical recipe for an engraved logo:

1. Drop your host part (a cube, a cylinder, an imported STL) — make sure it's **positive**.
2. Right-click its top face and pick *Apply texture to face…* from the context menu.
3. In the dialog: pick the pattern (try *voronoi* or *hex*), select **Negative** as the modifier, set *height* to your desired engraving depth (e.g. 0.6 mm), hit **Add to scene**.
4. The texture object lands flush against the picked face. Use the Inspector to nudge its position if needed.
5. Open the Slicer popover and hit **Slice & Export**. The G-code preview should show the texture as engraved channels on top of the host part.

Why the base plate matters

Every texture carries a small base plate (the **depth** parameter, 0.4–1.5 mm by default). This is intentional — without it, the relief geometry would float in air at the bottom and the CSG union/subtract could leave manifold-breaking gaps if your part's surface is slightly curved or non-coplanar with the texture.

The base plate gives the CSG engine some overlap to merge cleanly. For a positive texture, the base just embeds slightly into the host (no visual difference). For a negative texture, the base extends below the host surface so the engraved channels cut all the way through the reliefs.

Set *depth* to 0.4 mm as a minimum. Going lower can introduce manifold errors during slicing on highly curved hosts.

7. Walkthrough — knurled flashlight grip

End-to-end recipe combining everything above. Goal: a 25 mm diameter, 60 mm tall cylindrical grip with a fine diamond knurl.

1. From the Primitives tab, drop a **Cylinder**. Set radius = 12 mm, height = 60 mm.
2. Open **Library** → **Textures** (or Shift+T).
3. Pick **Knurl (Diamond)**. Set *tileSize* = 2 mm, *height* = 0.6 mm.
4. Switch **Wrap** from Flat to **Cylinder**. Leave radius at 0 (auto).
5. Set the texture's *w* to 2π × 12 = **75.4 mm** so it wraps exactly once. Set *d* to 60 (the grip's height).
6. Hit **Add to scene**. The knurled sleeve lands at the origin.
7. In the Inspector, position the texture object to coincide with the cylinder's centre line. The knurl visually merges with the cylinder body.
8. Open the Slicer popover, pick a printer, hit **Slice & Export**. Layer 1 should show the cylinder profile with the knurl reliefs around the perimeter.

> Save this as a Component (right-click → **Save to library**) once you're happy — you can reuse the knurled sleeve on future torches without rebuilding it.

8. Print-quality tips

Textures place additional demands on the printer because the reliefs are small features. Following these tips will save you from prints where the texture is blurry or missing entirely:

- **Layer height vs. relief height.** A 0.4 mm relief on a 0.2 mm layer is only 2 layers tall — the texture will read as flat. Keep relief *height* ≥ 3× layer height for crisp definition.
- **Nozzle diameter vs. tile size.** A 0.4 mm nozzle can resolve features down to ~0.5 mm. Tile sizes below 1.5 mm with a 0.4 mm nozzle will muddle. Use a 0.2 mm nozzle for the finest knurls.
- **Line width.** Set the slicer's outer wall width to match nozzle diameter, NOT 1.2×. Wider lines smear textures.
- **Ironing OFF on textured top surfaces.** Ironing flattens the relief. Disable ironing globally or use OrcaSlicer's per-object override.
- **Print speed.** Textured walls benefit from slower outer-wall speeds (20–35 mm/s). The reliefs ARE the outer wall.
- **Material choice.** PLA and PETG hold fine texture detail. ABS and ASA tend to smooth-over at typical print speeds due to higher viscosity and contraction.

9. Troubleshooting

Symptom	Likely cause	Fix
Texture is invisible in viewport	Wrong modifier — texture is negative but has no host to subtract from.	Add a positive primitive first, then apply the negative texture o
Slicer fails: NotManifold error	depth < 0.4 mm caused gaps in CSG union.	Bump depth to 0.5–0.8 mm and re-slice.
Print looks flat / no relief	Layer height too coarse for the relief height.	Either drop layer height to 0.12 mm or bump relief height to 1+
Wrap looks distorted	Width is wrong for target radius.	Use w = 2π × radius, or set wrapRadius directly and let w follo
Voronoi or hex_camo looks different every reload	Random seed not persisted with the project.	Save the project (Ctrl+S) — the seed is captured in the JSON.

10. Quick reference

Keyboard shortcuts: Shift+T opens Texture Library · Ctrl+D duplicates a selected texture · Ctrl+Z undoes a texture add.
Source: patterns live in `frontend/src/lib/textureGeometry.js`. Defaults are in `TEXTURE_PATTERNS` + `TEXTURE_DEFAULTS`.
Dialog: `frontend/src/components/dialogs/TextureLibraryDialog.jsx`.
Test: `frontend/tests/texture-geometry.mjs`.

Tutorial generated May 29, 2026 from ForgeSlicer source. Run `scripts/build_texture_tutorial.py` *to regenerate after future updates.*